BORNEO HACKWKND SUBMISSION REPORT

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PROJECT NAME : HORNBILL

CASE : AI FOR DISASTER RESILENCE

TRACK : NON-TECHNICAL

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1.0 EXECUTIVE SUMMARY

The Hornbill project introduces an offline-first emergency response ecosystem designed to bridge the "Macro-Deadzone" communication gap prevalent in rural ASEAN regions, such as interior Borneo. While traditional disaster response relies on a $4G/5G$ backbone that is prone to collapse during flash floods and landslides, Hornbill functions with 0% cellular connectivity through a Peer-to-Peer (P2P) mesh networking architecture. By utilizing on-device, lightweight AI for situational assessment and offline vector maps for navigation, Hornbill ensures that victims and first responders can coordinate life-saving logistics when the grid fails. This sustainable, B2G-focused solution directly aligns with UN-SDG 9 and 11, providing a resilient digital infrastructure for the most vulnerable populations in Southeast Asia.

2.0 INTRODUCTION

Problem Statement:

In the ASEAN region, over 5,000 disasters have been recorded since 2012, highlighting a critical need for enhanced preparedness. A fundamental failure point is the "Blackout Effect": during major flood or landslide incidents, cell towers are often the first infrastructure to be damaged or lose power. In rural regions like Borneo, characterized by dense forest and isolated terrain, it is economically and logistically difficult to install redundant hardware. This leaves communities digitally silenced and prevents responders (BOMBA/APM) from acting immediately due to a total lack of ground intel.

Case study:

AI solutions can strengthen resilience by providing hyper-local early warning systems and improving the mapping of vulnerable infrastructure. By utilizing AI for landslide detection and flood monitoring, the region can move toward a more proactive culture of safety as outlined in the **ASEAN Capacity Building Roadmap 2025-2030**.

Our Solution: AI for Disaster Resilience

Hornbill decouples emergency services from the internet.

- **Where AI comes in:** We use an on-device AI assistant to provide personalized survival advice and process visual data from photos to determine flood severity.
- **Intelligence at the Edge:** Instead of relying on cloud processing, Hornbill uses quantized AI models to convert user surroundings into lightweight JSON data packets that can be broadcast even when towers are offline.

3.0 SOCIAL IMPACT

Our project addresses the "Digital Divide" in disaster management by empowering three distinct social pillars.

PERSONA:

Name: Uncle Empari

Age: 54

Occupation: Community Leader & Rural Farmer

Context:

Uncle Empari owns a budget smartphone but lives in a "Macro-Deadzone." During the Northeast Monsoon, his community is frequently isolated by landslides that destroy local communication towers.

His Goals:

1. Send an SOS signal to nearby travellers or boatmen when towers are dead.
2. Receive guidance on how to secure his community's safety as waters rise.
3. Sync his community's medical needs with rescuers automatically once a signal "hop" is established.

Impact:

By using Hornbill, Empari regains his voice. He can use AI to verify flood levels and broadcast an SOS metadata packet that "hops" via BLE mesh networking to passing boatmen or drivers. This transforms him from an isolated victim into a Resilience Anchor for his entire longhouse.

Rescuer Efficiency (BOMBA & APM)

Traditional search and rescue in Borneo's dense canopy are often "blind" due to a lack of intel.

- **Impact:** Rescuers receive the **Automated SOS Metadata** generated by the victims' on-device AI. This provides precise GPS coordinates and a pre-triaged severity status (e.g., "Injured" or "Trapped").
- **Outcome:** This reduces response times in disconnected terrains and prevents first responders from entering "High Risk" zones without prior situational awareness.

Ensuring Educational Sustainability (Schools & Youth)

As noted in regional data, school-aged populations are among the most at risk during flash floods.

- **Impact:** Hornbill includes offline disaster preparedness training modules and automated "Survival Navigation" specifically designed for youth digital literacy levels.
- **Outcome:** By embedding resilience tools into students' devices, we ensure that even if separated from guardians, the next generation has the AI-guided tools to navigate to the nearest safe zone independently.

4.0 SDG Alignment:

SDG 9: Industry, Innovation, and Infrastructure (Target 9.1):

By using peer-to-peer mesh networking, we provide a reliable digital fallback that supports human well-being in areas where physical towers are economically or geographically unviable.

SDG 11: Sustainable Cities and Communities:

We aim to significantly reduce the number of people affected by disasters. By providing precise offline GPS coordinates and evacuation guidance to users, we reduce response times and protect vulnerable rural communities from the increasing risks of water-related disasters.

5.0 IMPLEMENTATION:

Hornbill's features are specifically designed for regions with poor or zero connectivity.

- **Smart Maps:** Uses pre-cached vector tiles (MBTiles) and the device's internal GPS hardware to find the nearest shelter and provide turn-by-turn navigation without fetching data from the cloud.
- **Bluetooth P2P Mesh:** Devices link directly using Bluetooth Low Energy (BLE). Messages "hop" from one device to another (e.g., from a victim to a passing boatman) until they reach a node with internet access. BLE is preferred over Classic Bluetooth because it doesn't require pairing and consumes low amounts of battery.

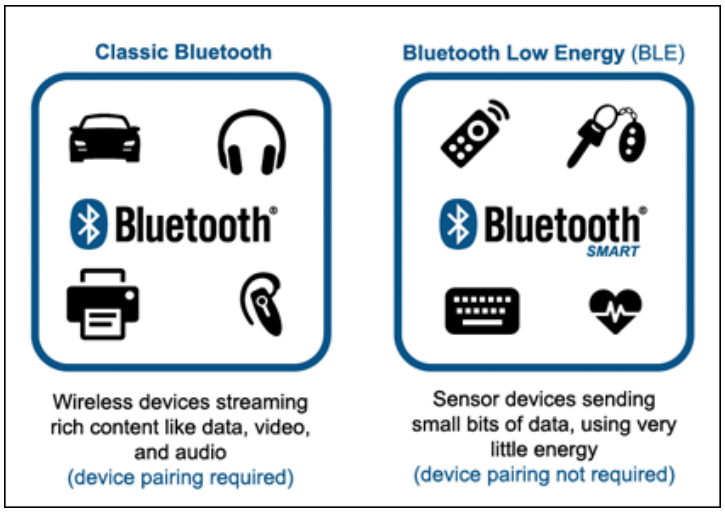

Figure 4.1: Classic Bluetooth vs. Bluetooth Low Energy (BLE)

- **AI Chatbot (Moondream):** A lightweight AI assistant provides information on what to do during specific disasters. It analyses surroundings via the camera to categorize damage severity and humanitarian needs.
- **Automated SOS Metadata:** AI converts images into JSON packages containing location, time, and status (e.g., "Active," "Injured," or "Unconscious") to minimize data size for mesh transmission.

AI ACKNOWLEDGEMENT

The Hornbill project utilized **Google Gemini (Pro/Vision)** for technical architecture refinement and metadata schema generation. On-device models were optimized using the **TensorFlow Lite Converter** to ensure performance on budget hardware. All technical decisions and intellectual content were validated and finalized by the human development team to maintain scholarly integrity.

6.0 TECH STACK:

Below is the tech stack for the proposed solution, alongside the reasoning behind each decision:

Framework: React Native

Provide cross-platform support on budget Android devices and full native control over BLE hardware.

AI Models: TensorFlow Lite (TFLite)

Useful for on-device, quantized vision models that are light enough to run without an NPU.

Databases: Firebase, SQLite

Firebase provides cloud synchronization when online, and SQLite for local storage of the "Digital Life Vault" and map data.

Maps: MBTiles

MBTiles format for efficient, offline vector rendering.

APIs: Malaysia Weather API, METMalaysia Web Service API

Contains open data for real-time regional warnings when a connection is available.

7.0 CHALLENGES FACED:

1. **BLE Connectivity:** Wi-Fi Direct was initially considered but proved too clunky. BLE mesh was chosen for its many-to-many capabilities, though it is limited to a ~100m range per hop.
2. **Hardware Constraints:** Running AI on budget devices requires a significant trade-off in model size. We optimized this through quantization to ensure the app remains under a manageable build size.
3. **Security:** Using **AES-CCM encryption** (the Bluetooth Mesh standard) ensures that sensitive medical data in the "Life Vault" is not compromised during P2P transmission.

8.0 FUTURE ROADMAP:

B2G Partnerships: Formalizing partnerships with **NADMA** and **METMalaysia** to secure disaster preparedness grants and integrate official predictive models. Integrate predictive data and run a live pilot in a high-risk rural district into every remote village prone to floods, landslides, and infrastructure failure.

Automated SOS Detection: Developing algorithms to detect flood inundation or sudden ground tilt automatically, triggering an SOS without user input.

Meteorological Crowdsourcing: Implementing feature to allow anonymized user reports in order to gather hyper-local data on flood patterns to improve future government prediction models.

Battery Optimization: Refining background BLE usage to ensure the mesh remains active for multiple days during extended power outages.

9.0 CONCLUSION:

Hornbill solves the "Last Mile" communication crisis by empowering rural communities to stay connected when the grid fails. By combining the social leadership of individuals like Uncle Empari with a decentralized, AI-driven mesh network, we move from reactive recovery to proactive resilience. Hornbill ensures that in the ASEAN region, no community is ever truly isolated.

10.0 REFERENCES:

1. ASEAN Capacity Building Roadmap 2025–2030 on Disaster Management: https://ahacentre.org/wp-content/uploads/2024/10/ASEAN-Capacity-Building-Roadmap-2025-2030-on-Disaster-Management-Spread-Version.pdf
2. UN Sustainable Development Goals (SDGs 9 & 11) Targets and Indicators: https://sdgs.un.org/goals
3. AHA Centre Annual Report 2024 (Regional Disaster Statistics and Impacts): https://ahacentre.org/publication/annual-report-2024/
4. Bluetooth SIG: Mesh Profile 1.0.1 Specification (Technical and Security Standards): https://www.bluetooth.com/specifications/specs/mesh-profile-1-0-1/
5. METMalaysia Open Data API (MET API Web Services): https://api.met.gov.my
6. Disaster Risk Reduction Knowledge Service (How AI is used to respond to natural disasters): https://ikcest-drr.data.ac.cn/post/lfa34
7. TensorFlow Lite: Model Optimization and Quantization for Mobile Devices: https://www.tensorflow.org/lite/performance/model_optimization